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Components sizing of photovoltaic stand-alone system based on loss of power supply probability

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ABSTRACT

The Stand-alone Photovoltaic System (SAPS) should be sized optimally since there no steady backup supply connected to it. An optimally sized SAPS should have a low overall cost without compromising the reliability of the system. This paper presents the review of the microgrid and the sizing of the SAPS. The review includes seven type of sizing methods for the microgrid. The sizing of the SAPS combines the Loss of Power Supply Probability (LPSP) and Life Cycle Cost (LCC) using the Iterative Method to determine the optimal size for the SAPS. The reliability of the SAPS is further improved using the Reliability Improvement Method (RIM). The location for the SAPS is at FKE Building, UTM, Johor. The results obtained using the RIM are compared with the results obtained from the LCC Method and the GCM. The GCM does not consider the overall cost of the SAPS throughout the lifetime of the system. The LCC Method consider the overall cost for the SAPS but the LPSP is fixed at 0.812%. While the proposed RIM improves the reliability of the SAPS up to 77.8% from 0.812% LPSP with only 4.9% cost increases.

1. Introduction

Nowadays, the fossil-fuel resources on a worldwide basis have been decreasing tremendously. A study was conducted on the fossil fuel usage in Bhopal, India shows that the oil, gas, and coal reserve will exhaust in 22 years, 30 years, and 80 years respectively. Therefore, it is necessary to find the alternative energy sources such as photovoltaic (PV) and wind energy. These energy sources have a huge potential to meet the continually increasing demand for energy since it is unlimited, pollution-free, and easy to access at any place. Therefore, the uses of renewable sources to fulfil the load demand in a remote location such as radio telecommunications, satellite earth stations, and rural areas are preferable [1,2]. Malaysia has the annual average of 4.21 kW h/m^2 to 5.56 kW h/m^2 of daily solar radiation [3]. The relatively high irradiance level throughout the year creates a potential for the stand-alone photovoltaic system (SAPS). The SAPS is both technically and economically practical for the remote area since the installation of the conventional grid system is not cost effective.

In general, the output power of these renewable sources are unpredictable and non-linear. Therefore, the energy storages such as the battery, the ultracapacitor, and the pumped hydro storage (PHS) are used for a short period of time to meet the peak load demand when there is a shortage of available energy to overcome the load demand [4]. The combination of the renewable energy sources with the energy storage can satisfy the source fluctuations. To optimise the utilisation of the renewable energy sources and the energy storage, a system sizing method is needed [5]. The sizing of SAPS is important in the system design since there is no steady backup supply connected to the system [6]. The optimum sizing can help to reduce the investment cost for installing the SAPS without sacrificing the reliability of PV system [1,5–8].

The optimum sizing that depends on reliability-cost relationship requires both of these components to have the measurable characteristic. The cost can be measured since it is a quantitative data. While the Loss of Power Supply Probability (LPSP) or the Day of Autonomy (DOA) are used to measure the system reliability. The LPSP is the ratio between the energy shortage and the total energy demands of the load for a long period of time [8]. It is also known as the Loss of Load Probability (LOLP), the Loss of Power Probability (LOPP), and the Load Coverage Rate (LCR) [9]. The value of the LPSP ranges from zero to one. A lower value of the LPSP shows a higher system reliability. The reliability results are summarised in a form of a sizing curve called isoreliability line which commonly consists of the sizes of PV arrays and the battery banks [9]. There are two approaches used in designing the SAPS using the LPSP, which are the Chronological Simulation and the Probabilistic Techniques [10]. The Chronological Simulation is done computationally and this method required data spanning for a period of time. The Probabilistic Technique uses the probability of the solar

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radiation and the load demand. The second technique is used if the solar radiation data is not available. The Chronological Simulation is a more precise method compared to the Probabilistic Technique.

There are several Chronological Simulation approaches to determine the optimum configuration for the SAPS. The Numerical Method is used to determine the size of the PV array and the battery story capacity for the SAPS using the hourly mythological data and load profile [11]. The LPSP and DOA concept are used to measure the reliability of the SAPS. The DOA uses the number of days for the SAPS to operate without any blackout [12]. Since Malaysia is not a seasonal country, the DOA is an inadequate concept to measure reliability. While the LPSP is a more suitable reliability measurement concept for the climate condition in Malaysia. The Graphic Construction Method (GCM) is another method for the sizing of the SAPS [1]. The optimum system's configuration is determined by finding the tangent between the partial derivative of a PV array and battery cost with the isoreliability line of the desired LPSP. However, this method does not consider the effect of the interest rate and inflation. A more advanced sizing method based on the LPSP concept is the Artificial Bee Swarm Optimisation (ABSO) for the Hybrid Photovoltaic System (HPVS) [13]. The ABSO is used to reduce the objective function for a complex system and implements the scholastic rules to identify a global solution. An Iterative Method is simple compared to the ABSO method. By using the Iterative Method, the reliability and the total cost of each system's configuration are calculated. A power based Iterative Method is proposed to determine the optimum size for the HPVS [14]. The objective is to determine the minimum different between the power generated and load demand. Another optimal sizing using Iterative Method combines the LSPS and the Levelised Cost of Energy (LCOE) [15,16]. The LCOE of the system's configurations with the desired LPSP is calculated and the lowest LCOE becomes the optimal size for the system.

This paper present the combination of the LPSP concept with the Life Cycle Cost (LCC) analysis to determine the optimal size for the SAPS. The Iterative Method is used to determine the LPSP and LCC of each configuration. The main objective of this work is to determine the system's configuration with a minimum LCC for the criteria of 1% LPSP. The system reliability is further improved by using the Reliability Improvement Method (RIM). This is to identify the maximum reliability improvement with minimum cost increment for the criterion of 1% LPSP. The results from the RIM and the LCC methods are compared with the results using the GCM. The simulation has been conducted using the input parameters from Senai, Johor, Malaysia. The data is based on the hourly solar irradiance and temperature, while the consumption of the FKE Building, UTM is used as the load profile. The simulation is conducted using MATLAB® simulation package. This article is organised as follow: introduction and a review of the sizing method for microgrid system, the available literature and related research in sizing methodology, followed by the system modelling and sizing in Section 3. Then, the results and discussions are described in Section 4 and lastly the conclusion.

2. Review of the component sizing for microgrid system

It is necessary to size the components of microgrid as this ensure the reliable power supply while remaining the economic viable of the system developed. The sizing of renewable energy in the microgrid is a complicated issue since the process of sizing needs to consider the reliability and the cost, which coupled to the uncertainty in demand requirement and energy production. In the microgrid that obtained the energy from renewable energy resources, the power generation depends on the weather conditions including the wind speed, the solar radiation and the ambient temperature. In a grid connected system, the sizing of the microgrid component is less complicated as it only considers the area and time duration to supply energy from the grid. The grid plays the role to ensure the energy demand is satisfied. In comparison, the stand-alone system has a more complex sizing system since it needs to work without disturbances and ensure the quality of power supply. The optimal design of a stand-alone microgrid system has to consider both reliability and cost issues [17]. This means the design of any renewable energy system should meet the load demand at a minimum cost whereby the design of the desired system can be evaluated through its lifetime and emission. The design of optimal size also has a high connection with the energy management strategy that ensures the power flow to meet the demand in the system. The lifetime cost in microgrid usually consists of the operational, the capital, and the maintenance cost. The capital and the maintenance cost are considered as the "fixed cost". However, the operational cost varies due to the adjustment in how the system works that could minimise the operational cost. The calculation of the lifetime cost and the changes in monetary value due to supply time need to be considered. Hence, the optimal design of the system must ponder the generators type along with the size that results in the lowest lifetime cost and gives a low emission level. In this regards, the goal of microgrid design process is to achieve a reliable power supply under varying atmosphere conditions with a minimum cost of the system. Due to the high requirement in sizing method for microgrid system, numerous articles in the literature discuss the technique used in developing the optimal size for microgrid system. The study on sizing for the stand-alone system seems to be more popular listed in the literature compared to grid connected system. The power management strategy for dispatching energy from the energy source, energy storage and electrical loads require a proper size of the hybrid component system. According to the previous literature, the sizing method for microgrid can be grouped into several techniques including the manually sizing method, the iterative sizing method, sizing using the commercial software tool and the sizing using the optimisation algorithm [18]. The method of manually sizing, sizing using software and optimisation sizing have been discussed by Behzadi in the article [19]. In addition to this three techniques, Upadhyay expanding the alternative to size the microgrid system in the article [20]. The alternatives are grouped into six methods including the Graphic Construction Method, the Probabilistic Method, the Analytical Method, the Iterative Method, the Artificial Intelligence Method and the Hybrid Method. Thus in this review section, the sizing method for the microgrid system is grouped into eight techniques as illustrated in Fig. 1. The description of each method with the related research work given in Table 1.

The manual sizing is being done in previous research by Chedid in [21] where the Analytic Hierarchy Process (AHP) quantify the parameters needed in designing the hybrid solar-wind power system (HSWPS). In this method, the trade-off risk method is used to generate multiple plans under 16 different futures and obtained the corresponding trade-off curves. There are several hierarchical sets up inside this method which includes all factors influencing the size of the developed system. This method is time-consuming since it requires manual calculation to obtain the size of the developed system. Another article by Chasheng in [22] where the researcher first considers the capacity factor of the renewable energy sources used. In this regards, the unit sizing aims to minimise the difference between the generated power (P_{gen}) from the renewable energy sources and the demand (P_{dem}) over a period of time. Although the sizing mechanism has reached the optimum target but the overall results are not feasible since it is not into account the cost effective factor. A manual size method for battery storage and super-capacitor is studied in article [23] where the corresponding calculation is properly done with the aim to increase battery lifetime. There are two methods used to calculate the size of the battery. First, consider the size of the wind turbine and second consider the size turbine and fluctuation generation of wind power due to the unstable wind speed. The mathematical equations for the proper size of the supercapacitor and the battery are available in literature [23] where all the calculations are account subjected to the State of Charge (SOC) of the supercapacitor and battery.

An Iterative Method is introduced in reference [24] where an iterative algorithm is described to determine the wind generator

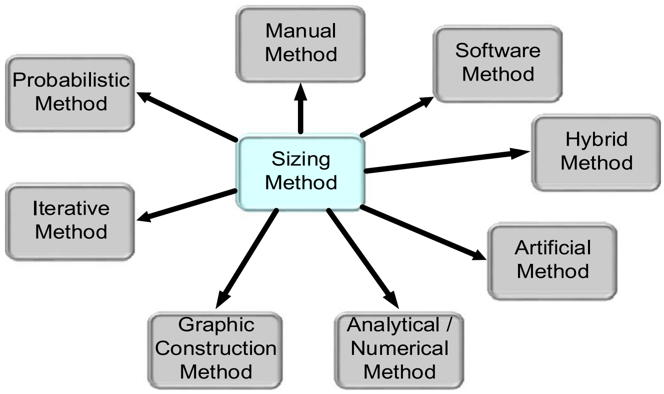

Fig. 1. Group of sizing method for microgrid system.

capacity and the number of PV panels needed for the standalone system. In this method, the algorithm uses the hourly average wind, insolation and power demand to obtain the wind and PV power generation capacities. The demand should also be full fill while minimising the annual cost to the customer. The iterative procedure performed through several steps including select commercial available unit sizes for the wind turbine, the PV panel and the battery storage. Next, the rating for the wind turbine is checked. Since the rating for the wind turbine is far exceeds a single PV panel, the number of turbines (K_w) is kept constant and the number of PV panels (K_s) is increased until the system is balanced. The calculation of system annual cost is done for each combination of K_w and K_s followed by the selection of the system's configuration with the lowest cost. Another researcher that used the Iterative Method in sizing their system is Kaldesis et al. where the method is described in [25]. The analysis provides a first estimation with also include the maintenance and the operation cost. However, this method is considered complicated since it provides a complete long-term energy production cost method of a wind-diesel hybrid system. Also, the fixed and variable costs of the initial cost, the maintenance cost, the operation cost, and the financing are taken into account for the analysis. Despite to this conventional techniques, Nikhil and Subhakar in [26] work with a new Iterative Method where they proposed an adaptive feedback iterative for a fast convergence to obtain the best optimum configuration of the PV panel and battery. In their work, a parametric analysis is carried out to examine the effect of load duration and charge controller low-voltage disconnect (LVD) on system sizing. Moreover, the authors in [27] have performed the sizing method by first design a conventional sizing model before they come out with the development of Iterative sizing algorithm (ISA). An Iterative-Pareto-Fuzzy (IPF) technique is used to obtain the optimum solution between all the design objectives. The results show that the inclusion of the unutilized surplus power criteria gave a different set of Pareto points to the system and provided a different result for the best-compromised solution. The total cost of the hybrid system is lower when the minimization of unutilized surplus power criteria is taken into account in the analysis [28].

For the new evolution of sizing method, the researcher started to use artificial intelligence and optimisation method to obtain the optimal design in their system. There are various commercially available software tools for sizing of the microgrid system are discussed in the literature. As previously mentioned, grid connected system seems to be easier to size compared to the stand-alone system as it relates to reliability and cost issues. Regarding this, the sizing of the stand-alone system will put the loss of power supply probability (LPSP) parameter as the main objective to be solved while remaining the minimum lifetime cost [17,29–33]. There are many discussions on the Optimisation Method to solve the sizing problems in the autonomous system. However, it is beyond of this manuscript to cover all of the available literature. Several works on Optimisation Method are presented in this section to explain the optimisation work for sizing an autonomous microgrid system. This is including a novel meta-heuristic algorithm based on Artificial Bee Swarm Optimisation(ABSO) is used to size the hybrid system consist of PV panels, WTs, and FC in article [13] where the simulation result give the PV/WT/FC is the most cost-effective hybrid system and WT/FC and PV/FC systems are in the other ranks. The Loss of Energy Expectation (LOEE) and the Loss of Load Expectation (LOLE) are studied in [34] where the objective function to achieve the minimum Cost of Energy (COE). Particle Swarm Optimisation (PSO) algorithm has been used in article [35] where multi-criteria objective function and multiple constraints are taken into account. While in [36], the PSO is used to reduce the Levelized Cost of Energy (LCOE) within an acceptable range of production. The Genetic Algorithm is a famous algorithm used in sizing the microgrid system which has a number of advantages including generally robust in identifying the global optimal solution, particularly in multimodal and multi-objective optimisation problems [37]. The sizing work in Refs. [5,19,38–42] use the GA to obtain the optimal size in their developed system. Basically GA work based on the evolutionary

Table 1
Sizing method.

Sizing Method	Sizing methodologies description	References
Manual Method	This method needs a hierarchical set up inside the chronology whereby all the factors that influence the system size should be included. Manual method requires manual calculation in obtaining the appropriate size for the system. It is time-consuming and not suitable for a large system.	[21,22]
Probabilistic Method	This is one of the simplest sizing methods. However, the results obtained may not be suitable to determine the best solution. In general, this method only requires a few parameters during the sizing process.	[54,55]
Artificial Method	Normally, artificial intelligence refers to the ability of a machine to perform similar functions that characterise human thought. Several algorithms including Genetic Algorithm (GA), Particle Swarm Optimisation (PSO) and Artificial Bee Colonies (ABC) are used to observe the sizing results.	[5,27,56,57]
Iterative Method	This method using the iterative performance evaluation where it is done by the recursive process. The iteration stop whenever the best configuration is achieved and reached the design specifications.	[25,26,40]
Graphic Construction Method	The sizing is performed based on the condition of the average value of the demand must be satisfied by the average values of the solar radiation and wind speed which represent a definite size of PV generators and wind turbine. The fundamental analysis of sizing curve is constructed between the available various sizes of PV generators and wind turbine.	[58,59]
Analytical Method	Normally, the desired hybrid energy systems are constructed by means of computational models. It describes the hybrid energy system size as a function of its reliability. This method requires less time compared to others method. In some case, it uses the software like HOMER to compute the design problems.	[60–63]
Software Method	The sizing method using the software like HOMER, HYBRID2 and IHOGA can give optimal sizing with provided several input parameters including load profile, solar radiation, wind speed and location.	[64,65]
Hybrid Method	This method combines at least two techniques which have the positive influence in obtaining the optimal result for the specific design objective. As long as the problem is multi objectives in nature, hybrid method can help to obtain the most significant result. However, a thorough understanding of each method has to be considered.	[66,67]

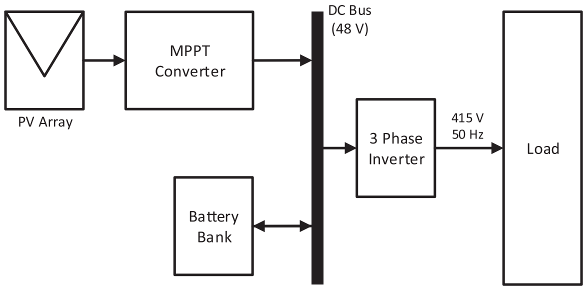

Fig. 2. Block diagram of the stand-alone PV system.

principle of natural genetics. It uses three operators namely selection, crossover and mutation to imitate the natural evolution process. The first step of a genetic evaluation is to determine the whether the chosen system's configuration which called a chromosome passes the functional evaluation, provides service to the load within the bounds set forth by the loss of power supply probability. If the evaluation is qualified, the chromosome has a lower Annualised Cost of System (ACS) than the lowest ACS value obtained at the previous iterations, the system's configuration (chromosome) is considered to be the best or optimal solution for the minimization problem in this iteration. This optimal solution is replaced by better solutions, if any, produced in subsequent GA generations during the program evolution. After the selection process, the optimal solution will next go to the crossover and mutation operations in order to produce the next generation population until a pre-specified number of generations have been reached or when a criterion that determines the convergence is satisfied.

Currently, there are various software tools are available in the renewable energy system market that can be used for the optimal design of the integrated system. Among all the available software tools, the Hybrid Optimisation Model for Electric Renewables (HOMER) is one the most popular tool for sizing of the integrated system. The HOMER is capable of modelling a power system's physical behaviour based on its Life Cycle Cost (LCC), which is the sum of installation and maintenance cost of system components over system lifetime. Besides, HOMER allows the programmer to compare many various design configurations on the basis of their technical and economic merits [43]. Many papers used HOMER in their sizing process can be referred in [34,44–51]. Another software utilised in the sizing methodology is HYBRID2 which it can define time intervals from 10 min to 1 h. In fact, the NREL suggest to first optimise the integrated system with HOMER and then proceed to improve the optimised system using HYBRID2. However, it needs long-term data for economic analysis of the integrated system. Other commercial tools used for sizing are TRNSYS, HYDRO GEMS, RETScreen, General Algebraic Modelling System (GAMS), Opt Quest, LINDO, WDILOG2, Simulation of Photovoltaic Energy Systems (Sim Pho Sys), Grid-connected Renewable Hybrid Systems Optimisation (GRHYSO), and H2RES. The descriptions of all these computational software methods are discussed in articles [52] and [53].

Although there are many methods can be used to size the microgrid system, but the type of generator installed and related constraint will reflect the selection if sizing method. Also, the number of components in the developed system must take into account before the method of sizing is selected. This is to ensure the impact of the chosen method is worth and give the significant result to the microgrid design objective.

3. Methodology of the stand-alone photovoltaic system modelling and sizing

3.1. System modelling

Fig. 2 shows the studied PV generation system to be studied for the LPSP simulation. It consists of the PV array connected to the MPPT converter. The MPPT converter is connected to a DC bus. The battery bank is connected to the bidirectional DC/DC converter for charging and discharging of the battery bank. Then, the three phase inverter is then connected to the load.

3.1.1. Photovoltaic arrays

The single diode PV model is used to calculate the power generated by the PV arrays, as shown in Fig. 3. To eliminate the five theoretical parameters such as the series resistance, R_s, and the ideality factor, A, the modified version of the model is used [68]. The hourly solar irradiance and temperature data are the input for the PV model. The PV module current, I_{pv}, is calculated using (1) to (3).

$$I_{pv} = I_{ref}\frac{I_{sc(ref)}}{I_{sc}} \tag{1}$$

$$I_{ref} = I_{sc}\left(1-k_{ref}^{\frac{V_{ref}+R_s I_{ref}}{V_{oc}}-1}\right) \tag{2}$$

$$V_{ref} = V_{pv} - (V_{oc(ref)} - V_{oc}) - R_s(I_{ref} - I) \tag{3}$$

where STC is the Standard Test Condition, I_{ref} is the reference current of photovoltaic model under STC (A), $I_{sc(ref)}$ is the short circuit current at reference condition (A), STC is the standard test condition at 1000 W/m^2 and 25 °C, I_{sc} is the short circuit current at STC (A), k_{ref} is the coefficient of k under STC condition, where V_{ref} is the reference voltage of photovoltaic model under STC (V), R_s is the series resistance of photovoltaic mathematical model, V_{pv} is the voltage of PV model (V), $V_{oc(ref)}$ is the open circuit voltage at reference condition (V), V_{oc} is the open circuit voltage at STC (V).

The output of the PV Array depends on the irradiance, G, and the PV module temperature, T. G and T affect I_{sc} and V_{oc}. I_{sc} and V_{oc} provided by the manufacturer only cover the measurement under the STC. Therefore, I_{sc} and V_{oc} under different G and T is calculated using (4) and (5) [68].

$$I_{sc} = I_{sc(ref)}[1 + k_{ti}(T - T_{stc})]\frac{G}{G_{stc}} \tag{4}$$

$$V_{oc} = V_{oc(ref)}\left[1 + k_{ti}ln\frac{G}{G_{stc}} + k_{tv}(T - T_{stc})\right] \tag{5}$$

where k_{ti} is the temperature coefficient of I_{sc}, k_{tv} is the temperature coefficient of V_{oc}, G_{stc} is the irradiance under STC (1000 W/m^2), and T_{stc} is the temperature under STC (25 °C).

The solar panel chooses for the project is the Sunpower SPR-315E-WHT-D (315 W) Solar Panel and the information of the PV panel is listed in Table 2 [69]. The PV panel is connected as 5 × 5 configuration (five PV modules is connected in series to become a PV string and 5 PV strings is connected in parallel to become PV array). The rated power produce by a single PV array is 7875 kW. The cost of the each PV module is RM 1356 and the cost of each PV array is RM 33,900, as shown in Table 3. The life cycle of the PV panel is expected to last for 25 years.

3.1.2. Battery bank

The PV arrays only able to generate electricity with the present of solar irradiance [70]. Therefore, energy storage system is typically needed to store and to supply the energy whenever the PV arrays fail to meet the load demand. The battery storage is one of the energy storage system used with the SAPS. The battery bank used for the work is East 220 V 10 kW EA9010II lead-acid battery pack. Each battery bank has 48 V rated terminal voltage and capable of delivering 10 kW of power. The State of Charge (SOC) for the battery bank is set to operate between 20% and 80%. This limit is set to increase the life expectancy of the battery. The cost of each battery bank is RM 4868 and the life cycle for the battery is seven years.

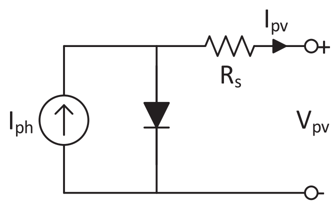

Fig. 3. Single diode photovoltaic circuit model.

3.1.3. Power converter

The SAPS consist of two types of power converters, namely the MPPT converter for the PV array and the three phase inverter. The MPPT converter is used in the SAPS because the PV array has a nonlinear I-V characteristic. The MPPT converter is also known as the solar charge controller. To allow maximum power generation by the PV array, the MPPT converter is added after the PV arrays. The MPPT converter determines the maximum power point of the PV arrays at given irradiance and temperature. There are several types of MPPT method used to maximise the output power produced by the PV arrays. The common MPPT method used in the PV generation system is the Perturb and Observe (P & O) method [71,72]. The P & O method is simple to implement. I_{pv} and V_{pv} from the PV arrays are measured and the PV arrays power, P_{pv}, is calculated by multiply V_{pv} with I_{pv}. The V_{pv} and P_{pv} are compared with the previous V_{pv} and P_{pv} respectively. The change of V_{pv}, V_{change}, and the change of P_{pv}, P_{change}, determine the perturbation direction. If the product of V_{change} and P_{change} is positive, the maximum power point voltage, V_{mppt}, increases. While if the product of V_{change} and P_{change} is negative, V_{mppt} decrease. The maximum power produced by the PV arrays according to the irradiance and temperature is determined using the P & O method.

The three-phase inverter is used to convert the 48 V DC power supply to 415 V three-phase AC power supply with the frequency of 50 Hz. The inverter chosen for the SAPS is the Leonics Apollo STP-2110Cp Parallel Inverter [73]. It has the power rating up to 18 kW and 10 units of this inverter can be connected simultaneously to a DC bus. For this work, four units of Leonics Apollo STP-2110Cp Parallel Inverter is connected to the DC bus. The efficiency of the MPPT converter and the three-phase inverter are set to 90%. While the cost of all the converters is RM 125,621, as shown in Table 3.

Table 2
Characteristic of the components in SAVS system.

Parameter	Value
Sunpower SPR-315E-WHT-D Photovoltaic Panel	
Rated power	315 W
Open circuit voltage	64.6 V
Short circuit current	6.14 A
Maximum power point voltage	54.7 V
Maximum power point current	5.76 A
Temperature coefficient of V_{oc}, k_{tv}	−176.6 mV/°C
Temperature coefficient of I_{sc}, k_{ti}	3.5 m mA/°C
PV array series-parallel connection	5 × 5
Life expectancy of the PV panel	25 years
Leonics Apollo STP-2110Cp Parallel Inverter	
Rated output power	4 × 18 kW
Rated output voltage	415 V
Output frequency	50 Hz
Efficiency	90%
Life expectancy of the Power Converter	13 years
East EA9010II Lead-acid Battery pack	
Power rating	10 kW
Voltage rating	48 V
Operating Level	20–80%
Battery Charging Efficiency	80%
Life expectancy of the battery bank	7 years

Table 3
The cost needed to determine the slope of the graph.

Parameter	Cost (RM)
Cost of a PV array, α	33,900
Cost of a battery bank, β	4868
Cost of power converter	125,621
Design cost and installation cost, C_0	150,000

3.2. System sizing

The component sizing process for the SAPS started with the solar irradiance, the temperature, and the load profile data, as shown in Fig. 4. The solar irradiance and the temperature data are converted to maximum power produced by a single PV array using the method in [74]. Then, the maximum power is used by the LPSP algorithm. The simulation of the LPSP is conducted for each number of battery banks and number of PV arrays. The simulation started with the system's configuration of one battery bank and one PV array. This system's configuration is analysed every hour in the duration of 365 days. After the analysis is completed, the LPSP for the system's configuration is calculated and stored in the first row and first column of the matrix. The similar simulation is conducted again with a different system's configuration of PV arrays and battery banks. The desired LPSP is chosen and the system's configuration that follows the desired LPSP is analysed to determine the optimum system's configuration. Three optimisation methods are chosen for this work, namely the GCM, the LCC method, and Reliability Improvement Method (RIM). The objectives of the optimisation are:

- The LPSP is equal to 1%
- The LCC is minimum
- Maximum reliability improvement with minimum cost increment from 1% LPSP

3.2.1. Input parameters description

The hourly load profile of FKE, UTM building during the weekday is shown in Fig. 5. The building consists of ten offices, six lecture rooms, five laboratories, and four restrooms. The main load for this building is air conditioning. The other loads include the lighting, the computer and the laboratory equipment. Fig. 5 shows the base load demand for the building is 16 kW and the peak load demand is 62.2 kW at 9 a.m. The load is at the lowest point between 11 p.m. and 6 a.m. The load begins to increase from base demand at 7 a.m. until 10 p.m. The load demand is the highest from 8 a.m. to 10 a.m. and 2 p.m. to 4 p.m. because the students are conducting the experiments in the laboratory. The load demand reduced to 42 kW at 11 a.m. to 12 p.m. because all the experiment in the laboratory has stopped. The 32 kW load between 5 p.m. and 10 p.m. is from the replacement classes conducted by the academic staff.

The hourly solar irradiance and temperature data are obtained from NASA Surface meteorology and Solar Energy [75]. The data available is in average for every hour in every month. The data location is at latitude 1.5° (Boundary between −1° and 2°) and longitude 103.5° (Boundary between 103° and 104°). By referring to Fig. 6(a), the data show the solar irradiance is at the highest in February and March and the lowest in December. The temperature is the highest on March (33.5 °C) and the lowest in December (30.9 °C) as shown in Fig. 6(b).

3.2.2. Loss of power supply probability algorithm

The unpredictable nature of the solar radiation influences the energy production of the SAPS. Therefore, the power reliability analysis is considered as an important step in designing PV system. There are various methods based on the reliability of supply available to determine the system's reliability. The reliability of the SAPS can be quantified using the LPSP and the recommended LPSP value for

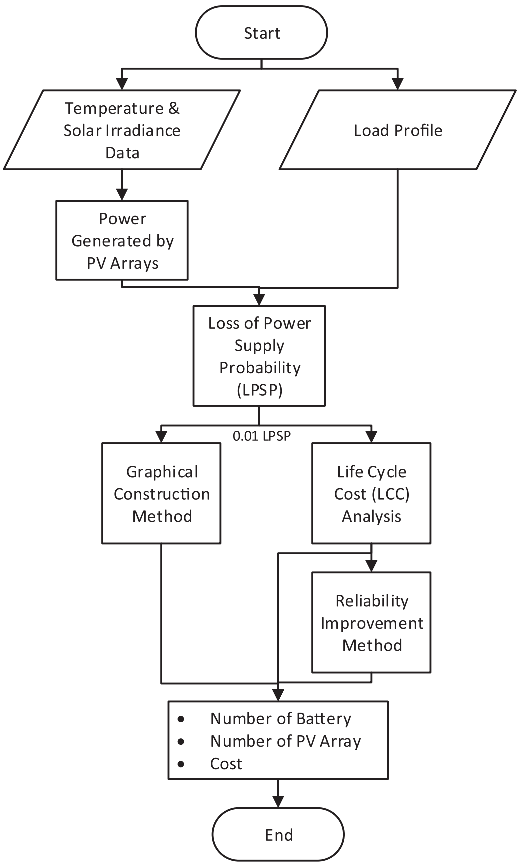

Fig. 4. The overview of the system component sizing.

domestic application is 10% [9]. The State of Charge (SOC) of the battery determines whether the battery is charging or discharging. To extend the battery life and preventing the battery from damage, the battery's SOC needs to be within the limit of B_{min} and B_{max}. When a high power is generated by PV arrays or a low demand exists, the battery may reach B_{max} and charge process needs to be stopped. While if the SOC reaches B_{min}, the load needs to be disconnected.

The Time-Series Simulation Method is commonly used in the renewable energy system optimisation with the meteorological data. The methodological data can be obtained from meteorological station data located near the SAPS. The PV system's behaviour is calculated based on one-hour interval meteorological input data. Borowy and Salameh [1] implements the time-series simulation to obtained LPSP using the GCM.

$$LPSP = Pr\ \{E_B(t) \le E_{B_{min}}; for t \le T \tag{6}$$

where E_B is the energy stored in battery banks at any time t and E_{Bmin} is the battery minimum allowable energy level.

There are two states for this model; the states where the energy generates by PV arrays exceed the load demand (State 1) and the state where the load demands are greater than the energy generated by the PV Cell (State 2). These two states are shown in Fig. 7.

For state 1, the equation is:

$$E_{B(t)} = E_{B(t-1)} + \left(E_{PV(t)} - \frac{E_{L(t)}}{\eta_{inv}}\right).\ \eta_{batt.in} \tag{7}$$

where $E_{B(t)}$ is the energy stored in battery banks in hour t, $E_{B(t-1)}$ is the energy stored in battery banks in previous hour, $E_{PV(t)}$ is the energy generate by PV module in hour t, $E_{L(t)}$ is load demand in hour t, η_{inv} is the efficiency of the inverter, and $H_{batt.in}$ is the round-trip efficiency of the battery banks.

The efficiency of charging and discharging of the battery banks is different. The manufacturer usually specifies a round-trip efficiency and the maximum depth of discharge. Since the battery capacity is defined as the energy extracted, the efficiency of battery discharging is 1.

For state 2, the equation is

$$E_{B(t)} = E_{B(t-1)} - \left(\frac{E_{L(t)}}{\eta_{inv}} - E_{PV(t)}\right) \tag{8}$$

The battery constraint is

$$E_{B_{min}} \le E_{B(t)} \le E_{B_{max}} \tag{9}$$

The condition called Loss of Power Supply, LPS, happened if the energy generated by PV arrays and the energy stored in the battery cannot fulfil the load demand. By referring to Fig. 7, the LPS can be expressed as

$$LPS_{(t)} = E_{L(t)} - (E_{PV(t)} + E_{B(t-1)} - E_{B_{min}}).\ \eta_{inv} \tag{10}$$

The loss of power supply probability (LPSP) for a period of T is the ratio of LPS with the sum of load demand [76] (Fig. 7).

$$LPSP = \frac{\sum_{t=1}^{T} LPS_t}{\sum_{t=1}^{T} E_{L(t)}} \tag{11}$$

3.2.3. Graphical construction method

The number of PV modules is the nonlinear function of the number of battery banks if the LPSP is kept constant. This relationship is shown in Fig. 8. It is important to determine the combination of PV/Battery's minimum cost that yields the needed LPSP. The optimum number of PV/Battery can be determined using the tangent of slope as shown in Fig. 8.

The cost function of the PV system is

$$C = \alpha.\ N_{PV} + \beta.\ N_{bat} + C_0 \tag{12}$$

where C is the capital cost of the system, α is the cost of PV module, N_{pv} is the number of PV module, β is the cost of a battery, N_{bat} is the number of battery banks, and C_o is the total constant cost (design cost and installation cost).

The optimum solution for function (2.12) is

$$\frac{\delta N_{PV}}{\delta N_{bat}} = -\frac{\beta}{\alpha} \tag{13}$$

3.2.4. Life cycle cost method

The LCC method is focused on analysing the cost involve during the lifespan of the system [77,78]. There are three considerations in calculating the LCC which are the initial cost, the maintenance cost, and the discount factor. The initial cost, C_{ini},is the cost needed to install the system. The annual maintenance cost, C_{main} is the cost needed to replace the wear and tear component after the system has been installed. The present factor, P, is the present value of cost that needs to be paid in the future. The present factor can be calculated using the equation [76,78–80].

$$P = F(1+i)^{-n} \tag{14}$$

where P is the present factor, F is the future cost, I is the interest rate, and n is the project life.

$$LCC = C_{ini} + (C_{main}.\ n.\ P) \tag{15}$$

where LCC is life cycle cost, C_{ini} is the initial cost and C_{main} is the annual maintenance cost.

4. Results and discussions

The results are divided into four parts which are the reliability, cost, reliability-cost relationship, and method comparison. The reliability

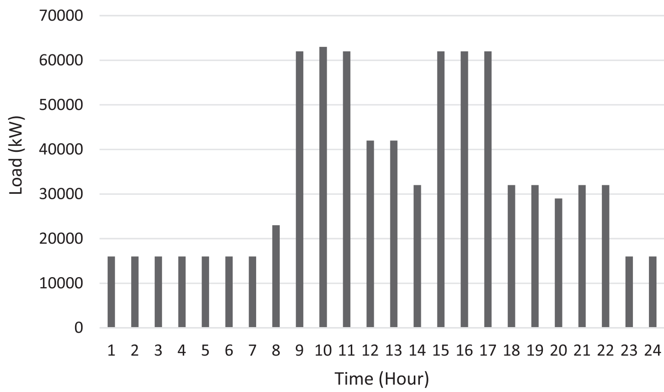

Fig. 5. Hourly load profile of FKE, UTM building.

simulation result shows the LSPS graph and the conversion of the LPSP graph into the isoreliability line. The cost simulation result consists of two different cost analysis methods which are the GCM and the LCC. The reliability-cost relationship improves the LPSP of the SAPS without significantly increase the LCC. The method comparison compares various PV sizing analysis with the proposed reliability-cost analysis to improve the SAPS.

4.1. Reliability simulation result

Fig. 9 shows the simulation results of the LPSP data. The data resolution is 300 × 300. It means that the simulation is started from one battery and one PV array (upper left-hand site of the graph) until 300 battery banks and 300 PV arrays (lower right-hand site of the graph). Therefore, there are 90,000 configurations of battery banks and PV arrays are tested to form Fig. 9. For each configuration, the simulation is conducted for every hour in 365 days. In each configuration, there are 8760 h of data processed. Each data is analysed to determine whether the supply is satisfied the load or unable to provide energy to the load. Therefore, there are 788.4 billion hours of data is analysed to obtain the LPSP graph shown in Fig. 9. The SAPS reliability is the ability of the stand-alone PV system to satisfy the load demand. The SAPS reliability for the proposed research work is measured using the LSPS. The supply never satisfied the load when the LPSP is one. While when the LPSP is zero, the supply will always satisfy the load. By referring to Fig. 9, the blue area has a low LPSP and a high SAPS reliability. While the red represents a high LPSP and a low SAPS reliability.

The LPSP is near to when the number of PV arrays is low even when the number of battery banks increases, as shown in Fig. 9. Since the battery bank is an energy storage and does not generate energy, increasing the number of battery banks does not reduce the LPSP of the SAPS if the PV arrays are not enough. This characteristic is shown in Fig. 9 as a horizontal dark red line that shows the LPSP of the system is one. This condition occurs when the number of PV arrays is one to four. During this condition, the PV arrays cannot satisfy the load demand. However, when the PV arrays are more than four, the LPSP begin to reduce. As a result, the contour begins to form and it is parallel to x-axis and y-axis. This contour represents the similar SAPS reliability by having the similar LPSP. This contour is called isoreliability line.

The isoreliability lines shown in Fig. 10 is derived from LPSP Graph shown in Fig. 9. These lines touch the system's configurations that have the same reliability. Since the SAPS reliability is measured as LPSP, all the system's configurations that touch the isoreliability line have the same LPSP value. The recommended value of LPSP for domestic application is 10% [9]. Therefore, the isoreliability line below 10% LPSP is extracted from LPSP graph. The review shows that 1% LPSP is more suitable for the SAPS [81].

The isoreliability lines for the SAPS shown in Fig. 10 consist of 10% LPSP, 1% LPSP, 0.1% LPSP, and 0.01% LPSP. The 1% LPSP, 0.1% LPSP, and 0.01% LPSP are 10 times, 100 times, and 1000 times more reliable comparing to the SAPS reliability with 10% LPSP respectively. The total load energy demand for a day and a year are 813 kW h and 296,745 kW h. Therefore, for the 10% LPSP, 1% LPSP, 0.1% LPSP, and 0.01% LPSP, the load demand that could not be satisfied is 29,674.50 kW h 2967.45 kW h, 296.75 kW h, and 29.68 kW h respectively.

The results in Fig. 10 show there are three different regions of the isoreliability line for every LPSP. The first region is the constant PV array region. In this region, the increase in the number of battery banks does not reduce the LPSP. Adding more battery into the system does not result in increases of the LPSP. The constant PV array region for 10% LPSP is 18 PV arrays and 50 battery banks and above. The constant PV array region for 1% LPSP is 20 PV arrays and 133 battery banks and above. The constant PV array region for 0.1% LPSP and 0.01% LPSP are 22 PV arrays with 141 and 187 battery banks and above respectively.

The second region is the constant battery region. In this region, the increases in the number of PV arrays does not reduce the LSPS, as shown in Fig. 10. This is due to the limitation of the energy storage which is the battery banks. The number of battery banks in the constant battery region for the 10% LPSP, 1% LPSP, 0.1% LPSP, and 0.01% LPSP are 38, 44, 128, and 174 respectively. While the number of PV arrays in the constant battery region for the 10% LPSP, 1% LPSP, 0.1% LPSP, and 0.01% LPSP are at least 49, 56, 43, and 46 respectively.

The third region is the optimum sizing region where the cost analysis for the SAPS sizing is conducted. The number of battery banks and PV arrays are not constant in this region. By referring to Fig. 10, the location of the optimum sizing region is relatively similar to the number of PV arrays. The location is between 18 and 56 PV arrays for four different LPSP shown in Fig. 10. However, the location of the optimum sizing region is different against the number of battery banks. As the LPSP decreases, the location of the optimum sizing region against y-axis becomes higher. This condition causing the constant PV array region to become similar and the constant battery region becomes far away.

This condition occurs because the PV arrays need to satisfy the load and charge the battery is enough when 18–56 PV arrays are used. However, the higher number of battery banks are needed to increase the reliability of the SAPS. The PV arrays only capable of producing the energy during the daytime. During night, the solar irradiance is absent and the load needs to rely on the energy obtained from the battery banks. However, the energy stored in the battery banks are limited. To increase the reliability of the SAPS, the number of battery banks needs to be increased. The increase in the number of battery banks allows a more energy stored and used during the night. Therefore, a higher energy stored, the lower the energy cannot be supplied to the load and reduced the LPSP. This leads to a higher LPSP requires a high number of battery banks and not a higher number of PV arrays.

4.2. Economical simulation result

There are two objectives when conducting the component sizing of the SAPS which are the reliability and the cost. The first objective is to determine the system's configuration that produced the desired reliability. The LPSP is chosen to quantify the SAPS reliability. The desired LPSP is chosen for the simulation is 1%. This shows that only 1% of the load demand is not delivered by the SAPS. The 1% LPSP is equal to 2967.45 kW h of energy shortage or roughly four days of power outage per year. The second objective is to determine the minimum cost for all the system's configuration that has a similar LPSP. Two methods are used to identify the minimum cost from the isoreliability line which are the GCM and the LCC method.

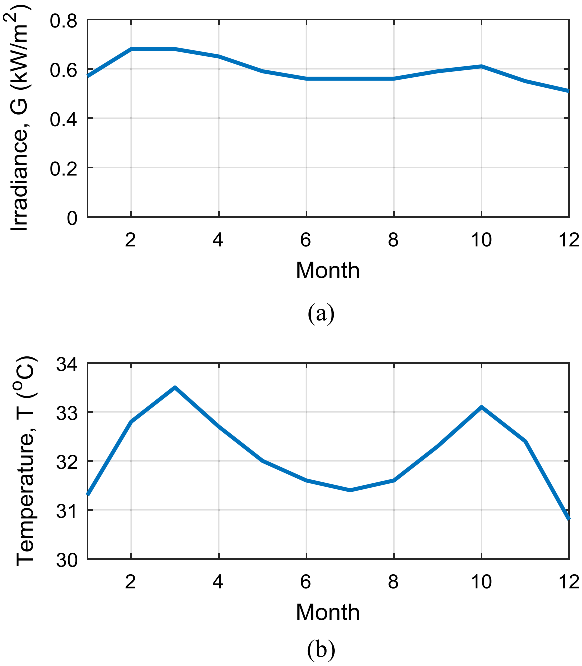

Fig. 6. Monthly averaged a) solar irradiance (kW/m^2) and b) temperature (°C) on a horizontal surface at 12 p.m.

4.2.1. Graphical construction method

All the system's configurations that intercept with the red line in Fig. 11 is having 1% LPSP. The next objective is to determine the cheapest configuration can be done so that the total cost of this work can be at its minimum point. The optimisation method based on Eq. (12) and Eq. (13) is shown in Fig. 11(a) and Fig. 11(b). The cost needed to determine the slope is shown in Table 3. From Eq. (12), the cost to build the system is RM 1,264,997. The slope is calculated using Eq. (13) and the result is −6.9638 and shown in Fig. 11 in the form of a blue line. The interception of the blue line with the 1% LPSP line is the optimum solution for the GCM. The optimum configuration for this method is 57 battery banks and 21 PV arrays.

The GCM is a simple method to determine the optimum system's configuration for the SAPS from the isoreliability line. The slope is based on the cost of the PV arrays and the battery banks. When the cost of a single PV array becomes higher compared to the battery banks, the slope becomes steeper. This ensures the blue line in Fig. 11(a) that intercept with the isoreliability line is more prone to a higher number of battery banks compared to the number of PV arrays in the optimum sizing region. Even though the GCM is simple to implement in the component sizing of the SAPS, the cost calculated does not cover the annual maintenance cost of the system. This is because the GCM only cover the cost of the PV arrays, the battery banks, the power converter, the design, and the installation. A more accurate cost calculation method is needed to determine the initial cost of the system as well as the overall cost of the system throughout the lifespan of the SAPS. One of the methods is the LCC.

4.2.2. Life cycle cost analysis

The disadvantage of using solely GCM is the cost analysis to determine the optimal configuration is the analysis is not considered the maintenance cost, salvation value and the interest rate. Therefore, the LCC method is used to determine the optimum system's configuration for the SAPS at given LPSP. In this subtopic, the LCC analysis is conducted on the 1% LPSP system's configurations.

Fig. 12 show the LCC of PV system's configuration that in contact with isoreliability line 1% LPSP. The x-axis of Fig. 12 refers to the configuration using the LCC method and the optimum size is configuration 177 which consist of 57 battery banks and 21 PV arrays. The LCC is the total of the initial cost and the total annual maintenance cost throughout the life of the components. The calculation of the annual maintenance cost will include the discount factor. The result of the simulation of the LCC is calculated using Eq. (14) and Eq. (15) and the LCC at the optimum configuration is RM 1,512,771. Fig. 12 show the relationship of the 0.01 LPSP system's configuration with the LCC of the SAPS. The relationship shows that as the number of PV arrays increases and the number of battery banks decreases, the LCC is increased. This is due to the high initial cost of the PV array. One PV array consists of 25 PV modules that cost around RM 1356 each. Therefore, the cost of one PV array is RM 33,900. Each battery only cost RM 4868. Therefore, a higher number of PV arrays results in a higher LCC. From Fig. 12, the most economic configuration is 177. The configuration 177 consist of 21 PV arrays and 57 battery banks. This configuration has the lowest PV arrays without compromising the reliability of the SAPS.

The LCC of the 176th configuration decreases significantly compared to the LCC of the 177th configuration. The LCC changes from RM 1,982,802 to RM 1,512,771. There are 125 battery banks and 20 PV arrays needed to achieve 1% LPSP at 176th configuration. However, when one PV array is added, the battery banks needed to maintain 1% LPSP drop significantly. The results show in the 177th configuration, 21 PV arrays and 57 battery banks needed to achieve 1% LPSP. The 78% reduction in the number of battery banks reduced the LCC significantly. This result shows that by adding one PV array into the system, RM 470,031 is saved without sacrificing the reliability of the SAPS.

4.3. Reliability improvement method

The desired LPSP is commonly fixed for to determine the optimum size for the SAPS. However, the fixed LPSP removes the possibility of improving the SAPS reliability without significantly increases the LCC. By referring to Fig. 10, the relationship between the LPSP, the number of battery banks, and the PV arrays are not linear. The optimum sizing region between 10% LPSP and 1% LPSP is very close. Only a small number of battery banks are needed to increase the LPSP from 10% to 1%. The number of battery banks in the optimum sizing region for the 10% LPSP is 38–50. While the number of battery banks in the optimum sizing region for the 1% LPSP is 44–61. However, the optimum sizing region for 1% LPSP and 0.1% LSPS is very far apart. The number of battery banks in the optimum sizing region for the 0.1% LPSP is 128–141. This nonlinear relationship of the LPSP with the system's configuration shows that there is a possibility of improving the SAPS reliability without significantly increase the LCC.

Fig. 13 shows the LCC of the optimum configuration at given LPSP. For this analysis, only the LCC method is used while the GCM is neglected since the LCC represent the overall cost of the stand-alone system throughout the system lifetime. The relationship shows that the LCC of the system will be higher if the LPSP is lower. This shows that the cost of the SAPS becomes higher if a more reliable system is needed. The increase in the LCC is due to the increases in the number of PV arrays and battery banks to follow the desired LPSP. However, the increase in the LCC against the LPSP is not linear. By referring to Fig. 13, there is a slight increase in the LCC when the LPSP decrease from 1% to 0.18%. The LCC only increase from RM 1,512,771 to RM 1,669,170. However, as the LPSP decrease further from 0.18%, the LCC significantly increases. The LCC increases from RM 1,669,170 to RM 2,176,280. Therefore, the Reliability Improvement Method (RIM) reduced the LPSP based on the significant of the LCC increases using Eqs. (16) and (17).

The original LPSP is 1%. An analysis was conducted to determine the value of improved LPSP that gives a small increase in LCC. The improved LPSP and the increased LCC is compared with 1% LPSP and the corresponding LCC (RM 1,512,771). The reliability improvement and cost increases are calculated using Eq. (16) and Eq. (17). The result of the analysis in shown in Fig. 14.

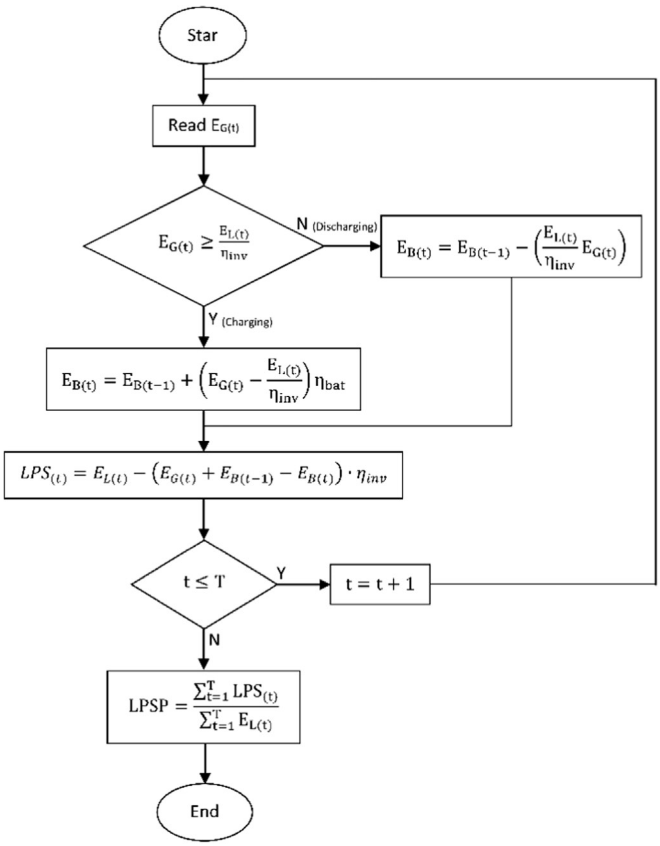

Fig. 7. The flowchart for the LPSP [1].

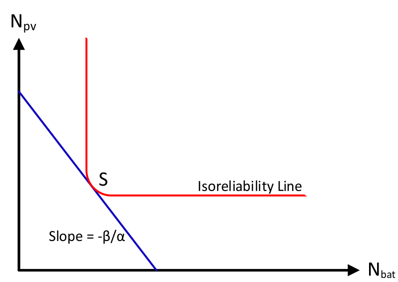

Fig. 8. Number of PV modules vs. number of battery banks at given Loss of Power Supply Probability (LPSP).

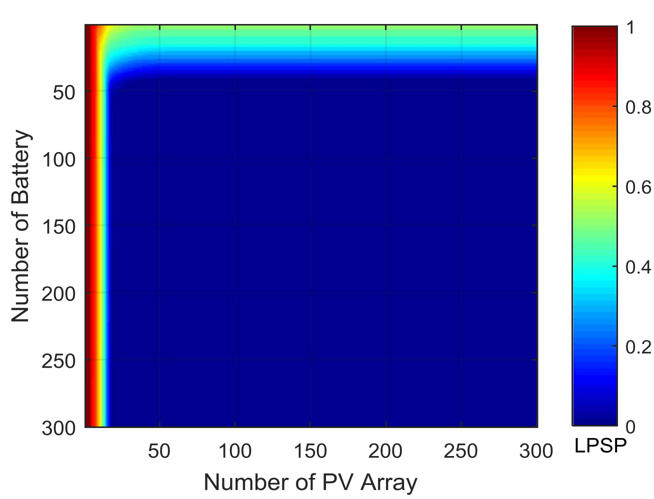

Fig. 9. Loss of Power Supply Probability (LPSP) simulation result for 90,000 configurations of PV arrays and battery.

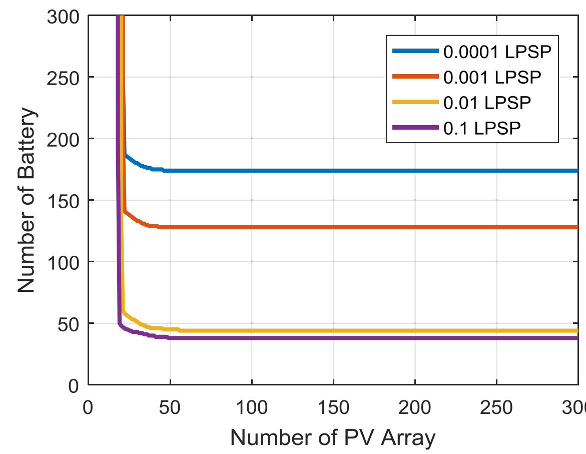

Fig. 10. Isoreliability line.

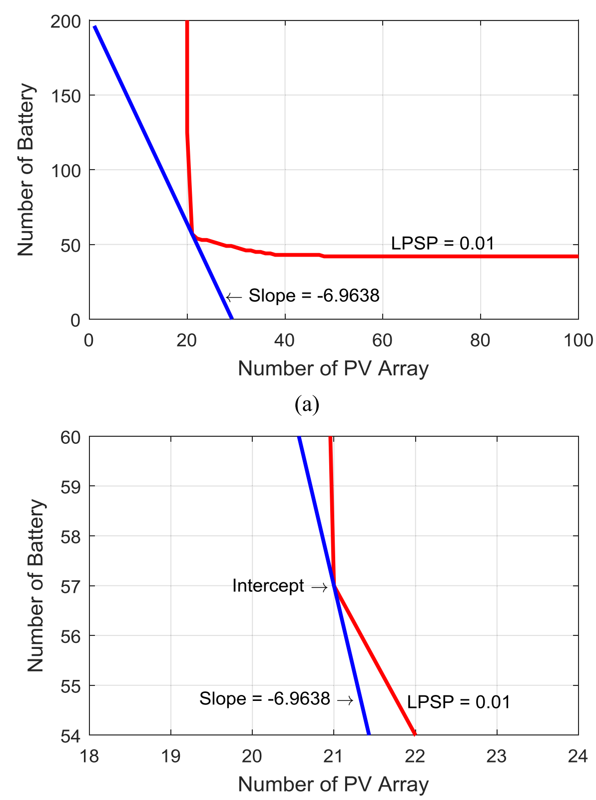

Fig. 11. An optimal solution using Graphical Construction Method (GCM). a) Overall view. b) Zoomed-in view of (a): intercept point between 1% LPSP and the slope of −6.9638 (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.).

$$ReliabilityImprovement = \frac{OriginalLPSP - ImprovedLPSP}{OriginalLSPS} \times 100\% = \frac{0.01 - LPSP_{Imp}}{0.01} \times 100\% \quad (16)$$

$$CostIncrease = \left|\frac{LCCof1\%LPSP - LCCofImprovedLPSP}{LCCof1\%LPSP}\right| \times 100\% = \left|\frac{1.512.771 - LCC_{LPSPImp}}{1.512.771}\right| \times 100\% \quad (17)$$

By referring to Fig. 14, changing the LPSP from 1% to 0.18% can improve the SAPS reliability up to 81.89% and the LCC only increases

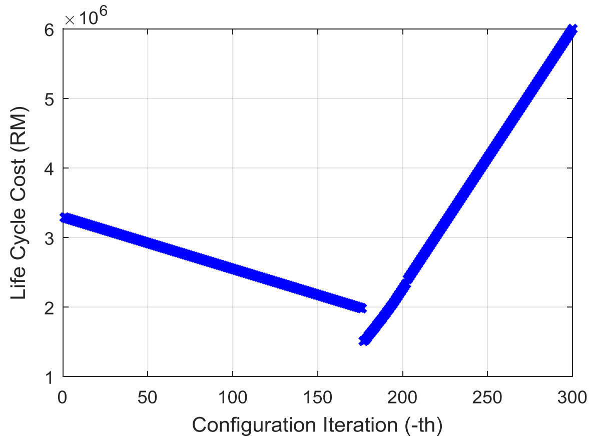

Fig. 12. Life Cycle Cost (LCC) corresponding to the configurations iteration.

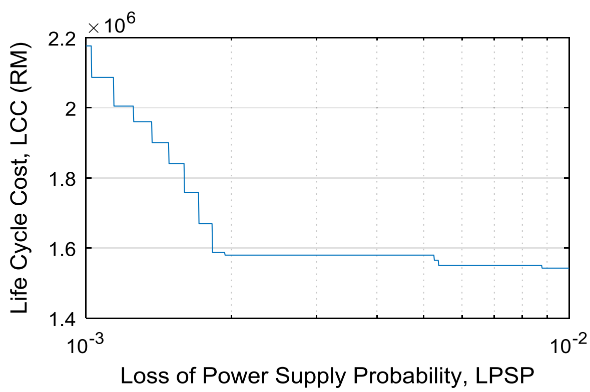

Fig. 13. Relationship between Loss of Power Supply Probability (LPSP) and Life Cycle Cost (LCC).

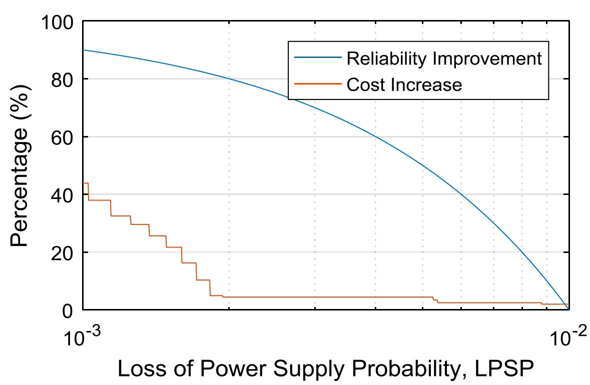

Fig. 14. Effect of reliability improvement on the increment in the Life Cycle Cost.

4.916% (RM 1,512,771 to RM 1,587,139). This shows that the SAPS reliability can be improved significantly without significant increases in the LCC. As a result, the Loss of Power Supply, LPS, reduces from 2967.45 kW h to 534.14 kW h with just 4.916% increases in the LCC. The number of PV arrays and battery banks for 0.18%LPSP with the minimum LCC are 22 and 62 respectively. The further decrease in LPSP value will result in a tremendous increase in the LCC. To increase the LPSP from 0.18% to 0.17% (1.14% reliability improvement), the LCC increases 5.424%. However, the increases in the LPSP from 1% to 0.18% improve the SAPS reliability up to 81.89% with only 4.916% increases in the LCC.

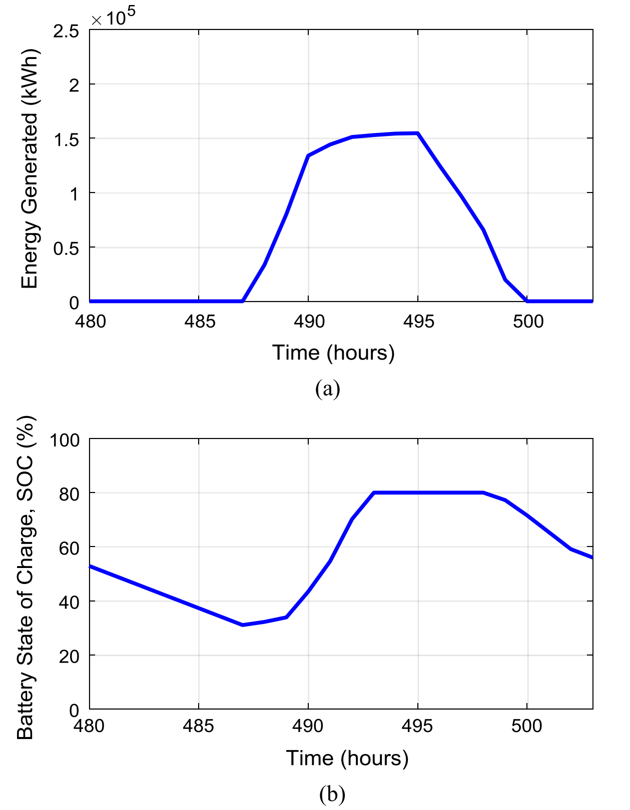

Fig. 15. Data sample for a) energy generated by 22 PV arrays and b) state of charge of 62 battery banks on the 20th day of the simulation.

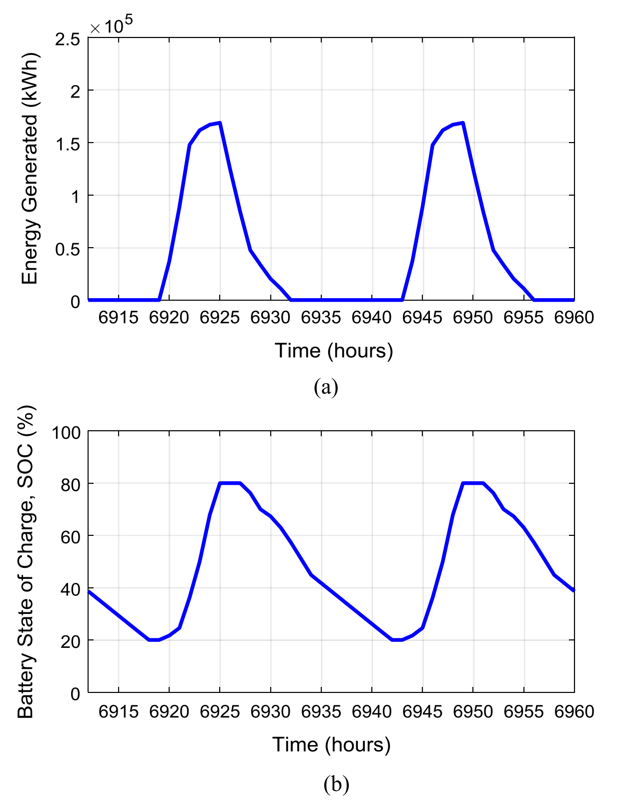

Fig. 16. Data sample for a) energy generated by 22 PV arrays and b) state of charge of 62 battery banks on the 288th day of the simulation.

Table 4
Comparison between Graphical Construction Method (GCM) and Life Cycle Cost (LCC).

Sizing Method	Graphical Construction Method	Life Cycle Cost	Life Cycle Cost + Reliability Improvement Method
Number of Battery Banks	57	57	62
Number of PV Arrays	21	21	22
Actual LPSP	0.812%	0.812%	0.18%
Initial Cost (RM)	1,264,997	1,264,997	1,323,237
Annual Maintenance Cost (RM)	Not Considered	247,774	263,902
Initial Cost + Annual Maintenance Cost (RM)	1,264,997	1,512,771	1,587,139

4.4. Energy generation and state of charge

Fig. 15(a) and Fig. 15(b) shows the data sample collected from the LPSP simulation using 62 battery banks and 22 PV arrays on the 20th day of the simulation. The result shows that a high energy generated by PV arrays during daytime since there is the present of solar irradiance. The maximum and total energy produced during the 20th day are 154.5 kW h and 1310.9 kW h respectively. At night, the energy generated by PV arrays are zero since there no solar irradiance present. Since battery level is set at a minimum of 20% to increase the lifespan of the battery, it is assumed the battery is depleted if the level reaches 20%. By referring to Fig. 15(a), the battery level never reaches 20% limit. Therefore, during the 20th day of the simulation, there no supply outage occur. The battery level only reaches 31.3% at 7 a.m. The battery is fully charged if the battery level reaches 80%. During the 20th day of the simulation, the battery is fully charged at 1 p.m.

The energy generated by 22 PV arrays and the state of charge of 62 battery banks on the 288th day of the simulation are shown in Fig. 16(a) and Fig. 16(b) respectively. The maximum and total energy generated by 22 PV arrays on the 288th day are 168.6 kW h and 1090.6 kW h respectively, as shown in Fig. 16(a). However, the energy generated during the 288th day is not enough for the energy storage of the 62 battery banks. Subsequently, the battery depleted on the 289th day at 5 a.m. since the battery level reaches 20%. At this point, the load experienced supply outage. The total of 32 kW h of energy demand is not delivered on the 289th day.

4.5. Method comparison

There are three optimisation methods used in this paper to determine the optimum size for the SAPS, as shown in Table 4. These methods include the GCM, the LCC method, and the combination of the LCC method with the Reliability Improvement Method (RIM). Even though the LPSP is set to 1%, the LPSP for each configuration could not be exactly 1%. This is because the configuration needs to be in the whole number since each PV array and battery bank cannot be partially installed. Hence, the GCM and the LCC method produce 0.812% LPSP instead of the 1% LPSP.

The GCM and the LCC method produce a similar system's configuration which consists of 57 battery banks and 21 PV arrays. Therefore, the SAPS reliability for both optimisation method is 0.812% LPSP, as shown in Table 4. However, the cost for both optimisation methods is different. The GCM considers only the initial cost of the SAPS. This initial cost includes the PV arrays, the battery banks, the power converter, the design, and the installation. However, the LCC is higher than the cost when using the GCM since the LCC includes the cost of the SAPS throughout the lifetime. The additional cost is the RM 263,902 annual maintenance cost for the system. The LCC method allows the analysis of the cost for the system to be done more accurately compared to the GCM. The RIM uses 22 PV arrays and 62 battery banks. This shows the RIM uses more battery banks and the PV arrays compared to the GCM and LCC methods which leads to a more reliability system. The SAPS reliability improves from 0.81% LPSP to 0.18% LPSP. Even though the SAPS reliability is significantly increased, the cost is slightly increased. There is 77.8% SAPS reliability improvement with only 4.9% LCC increases.

This research work is compared with another two researches in the literature, as shown in Table 5 [81,82]. The compared research works use the LPSP and the LCC method to assess the reliability of the SAPS. However, Research Work 1 includes Levelized Cost of Energy (LCOE) for assessing the economical factor of the system. The LCOE is the average cost per kW h of the energy produced by the system throughout the system operation. A lower LCOE is desired since the cost is cheaper energy generation. The proposed research work uses two different optimisation method. This includes the GCM and iterative method. While Research Work 1 uses only the iterative method. Research Work 2 uses Design Space Approach to determine the optimal sizing. The Design Space Approach allows multiple objectives consideration in the optimisation. This method includes the control of the LPSP, excess energy, capital cost, and the LCC of the system. Subsequently, two different optimisations solution is obtained. One of the solutions has a high reliability with a high cost (0.2% LPSP and RM 24,984 LCC). The other solution has a low LCC with a low cost (1% LPSP and RM 23,123 LCC).

The Iterative Method alone does not include the excess energy constraint as well as the improved reliability option. However, the proposed research work combines the Iterative Method and the Reliability Improvement Method. This allows a high improvement in the SAPS reliability without highly affects the LCC. This leads to the SAPS reliability improvement up to 77.8% from 1% LPSP with only 4.9% LCC increases. Research Work 1 and Research Work 2 use the SAPS for the house in the rural area and the residential lighting respectively. The load in Research Work 1 is only 2 kW h a day and the load in this research work is 813,000 kW h a day. This is because the load in the academic building is very high due to the usage of the air

Table 5
Comparison of the component sizing method using LPSP for the Stand-alone PV System.

Improvement	Proposed Research Work	Past Research Work:	
		Research Work 1 [81] (2016)	Research Work 2 [82] (2015)
Economical Assessment	LCC	LCC and Levelized Cost of Energy (LCOE)	LCC
Optimisation Method	GCM & Iterative Method	Iterative Method	Design Space Approach
Optimisation Objectives	Maximum LPSP improvement from 1% LPSP with low LCC increment	1% LPSP with minimum LCC	1% LPSP, Less than 30% excess energy, less than RM 20,000 capital cost, and less than RM32,000 LCC.
Reliability Optimisation	Yes	No	Yes
Actual LSPS	0.81% and 0.18%	1%	1% and 0.2%
Load	Academic Building	House in Rural Area	Residential Lighting
Daily Load Demand	813,000 kW h	2 kW h	n/a
LCC	RM 1,512,771 and RM 1,587,139	RM 20,752	RM 23,132 and RM 24,984

conditioning system. As a result, the LCC is very different. The LCC for Research Work 1 and Research Work 2 is around RM 20,000 to RM 30,000. While the LCC for the proposed research work is around RM 1,500,000 to RM 1,600,000.

5. Conclusion

The case study of this work is to design a SAPS for FKE Buildings, UTM. The simulation used one year of the meteorological data combined with the load profile of the FKE Building on a weekday. By using MATLAB®, the SAPS reliability for different configuration of the system is calculated using Lost of Load Probability by varying the number of battery banks and PV arrays. The LPSP is set to the acceptable reliability of the case study which is 1%. There are two sizing optimisation methods used in the study which is the GCM and the iterative method. Using the GCM, the optimal configuration is determined which is 52 battery banks and 21 PV arrays. The initial cost is RM 1,264,997. A more accurate sizing optimisation is done using the LCC method to determine the overall cost of this work throughout its lifespan. The Iterative Method has been used to determine the optimum size for the SAPS. The results show similarity with the GCM with the increase in the total cost due to the additional maintenance cost. The LCC for the SAPS is RM 1,512,771. The Iterative Method is a simple sizing optimisation method. However, this method requires a long simulation time since all the system's configurations are analysed. The SAPS reliability is further improved by analysing the changing rate of the LPSP and LCC. This reliability improvement analysis can be conducted because all the system's configurations are analysed by the Iterative Method. The simulation shows the SAPS reliability can be further improved up to 77.8% from 1% LSPS with an only 4.9% increases in the LCC.

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